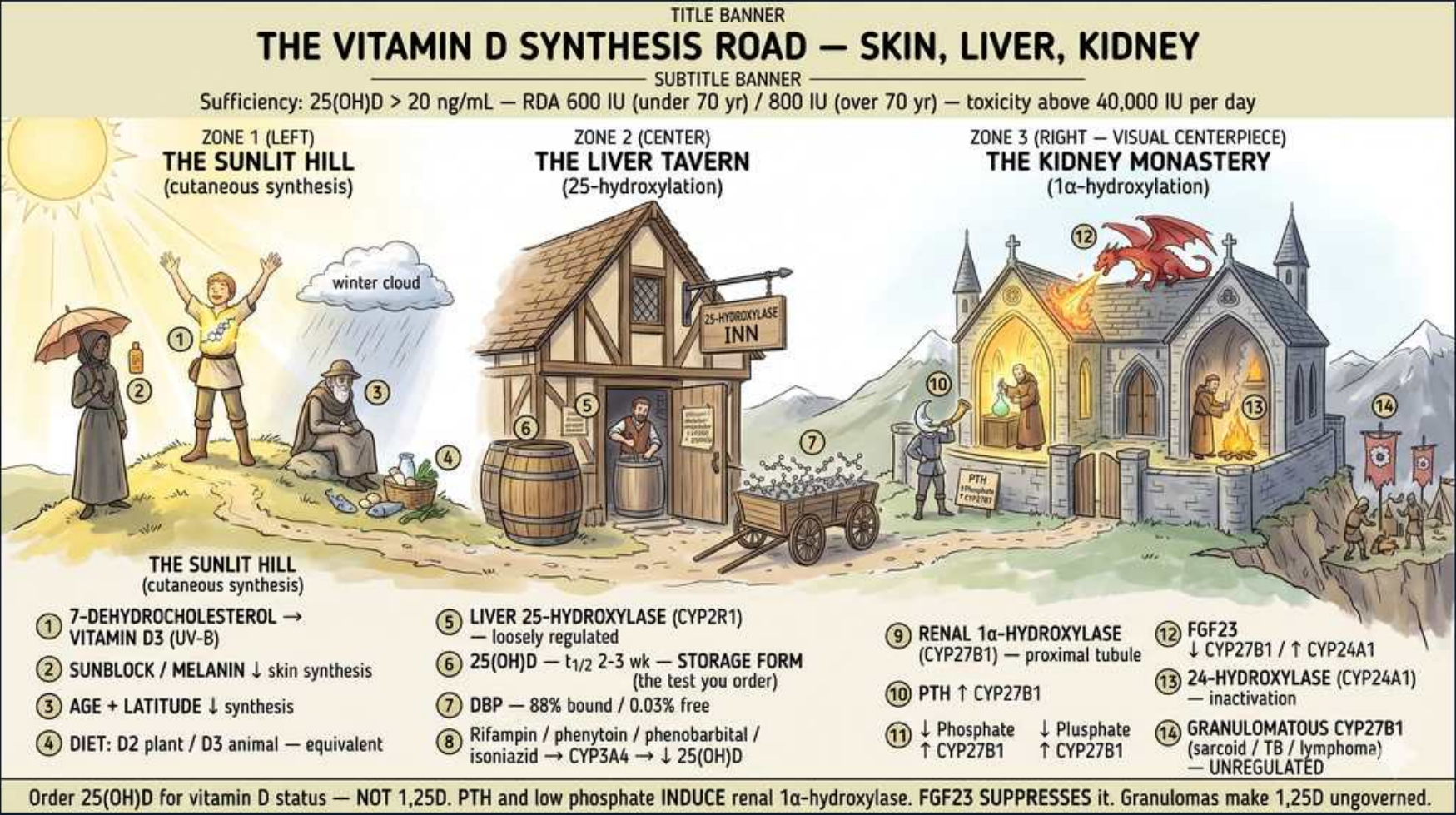


Vitamin D Synthesis — Skin, Liver, Kidney Activation & FGF23 Control



1. 7-DHC → vitamin D3 (UV-B)

WHAT YOU SEE

Sunlit hill, sunbather

WHAT IT ENCODES

UV-B light converts 7-dehydrocholesterol in skin to cholecalciferol (vitamin D3). This is the cutaneous synthesis step and the body's main source when sun is adequate.

BOARD TRAP

Glass/sunscreen/winter latitude block UV-B → relying on diet alone. D3 (animal) and D2 (plant/ergocalciferol) are roughly equivalent but D3 raises 25-OHD more reliably.

2. Sunblock/melanin ↓ synthesis

WHAT YOU SEE

Umbrella, dark-veiled figure

WHAT IT ENCODES

Melanin and sunscreen absorb UV-B and reduce cutaneous vitamin D synthesis. Darker skin, veiling, institutionalization, and high latitude all predispose to deficiency.

BOARD TRAP

This is why elderly nursing-home and veiled patients present with osteomalacia/secondary hyperPTH despite living in sunny climates.

3. Age + latitude + season

WHAT YOU SEE

Winter cloud / sun-angle icon

WHAT IT ENCODES

Cutaneous synthesis falls with age (less 7-DHC), higher latitude, and winter (low UV-B angle). These compound dietary inadequacy.

BOARD TRAP

A low 25-OHD in winter at high latitude can be physiologic seasonal nadir — interpret with PTH and Ca, not in isolation.

4. Diet: D2 plant / D3 animal

WHAT YOU SEE

Plate of food

WHAT IT ENCODES

Dietary vitamin D: D2 (ergocalciferol, plant/fortified) and D3 (cholecalciferol, animal). RDA ~600 IU (<70 yr) / 800 IU (>70 yr); toxicity above ~40,000 IU/day chronically.

BOARD TRAP

Sufficiency is 25-OHD >20 ng/mL. Toxicity causes hypercalcemia with SUPPRESSED PTH and high 25-OHD but NORMAL/low 1,25 — overdose is at the storage form, not calcitriol.

5. Liver 25-hydroxylase (CYP2R1)

WHAT YOU SEE

Liver Tavern / barrels

WHAT IT ENCODES

The liver performs 25-hydroxylation (mainly CYP2R1) to make 25(OH)D — the STORAGE form and the test you order to assess vitamin D status. This step is loosely regulated.

BOARD TRAP

ORDER 25(OH)D, NOT 1,25(OH)2D, to assess deficiency. 1,25 can be normal/high in deficiency due to compensatory secondary hyperPTH — a common ordering trap.

6. 25(OH)D — t½ 2–3 wk storage

WHAT YOU SEE

Barrel labeled storage

WHAT IT ENCODES

25(OH)D has a half-life of ~2–3 weeks, making it the stable status marker. It is ~88% bound to DBP, ~0.03% free.

BOARD TRAP

In nephrotic syndrome or liver disease, low DBP lowers TOTAL 25-OHD but free may be adequate — interpret with caution.

7. DBP — 88% bound

WHAT YOU SEE

Transport-protein icon

WHAT IT ENCODES

Vitamin D-binding protein carries ~88% of circulating 25(OH)D; ~0.03% is free. Free hormone is the bioactive fraction.

BOARD TRAP

Pregnancy and OCPs raise DBP → high total 25/1,25-D with normal free — mirrors the TBG effect on thyroid hormone interpretation.

8. Rifampin/phenytoin ↑ catabolism

WHAT YOU SEE

Pills accelerating breakdown

WHAT IT ENCODES

CYP enzyme inducers (rifampin, phenytoin, phenobarbital, isoniazid) accelerate vitamin D catabolism (CYP3A4/CYP24A1) → drug-induced osteomalacia.

BOARD TRAP

Chronic antiepileptic use is a classic board cause of low 25-OHD and osteomalacia — supplement these patients.

9. Renal 1 α -hydroxylase (CYP27B1)

WHAT YOU SEE

Kidney Monastery, dragon

WHAT IT ENCODES

The kidney proximal tubule performs the ACTIVATING 1-alpha-hydroxylation (CYP27B1) of 25(OH)D $\rightarrow$ 1,25(OH)₂D (calcitriol), the active hormone. This is the tightly regulated rate-limiting step.

BOARD TRAP

In renal failure, loss of CYP27B1 $\rightarrow$ low calcitriol $\rightarrow$ hypocalcemia $\rightarrow$ secondary hyperPTH (CKD-MBD). Treat with active calcitriol/analog, not plain D alone.

10. PTH $\uparrow$ CYP27B1

WHAT YOU SEE

PTH arrow boosting kidney enzyme

WHAT IT ENCODES

PTH and LOW phosphate INDUCE renal 1-alpha-hydroxylase $\rightarrow$ more calcitriol. This closes the loop: low Ca $\rightarrow$ PTH $\rightarrow$ activate vitamin D $\rightarrow$ raise gut Ca absorption.

BOARD TRAP

PTH stimulates 1,25 production; calcitriol then feeds back to suppress PTH — the basis of using calcitriol/calcimimetics in secondary hyperPTH.

11. $\downarrow$ Phosphate $\uparrow$ activation

WHAT YOU SEE

Low-phosphate signal to kidney

WHAT IT ENCODES

Hypophosphatemia independently stimulates 1-alpha-hydroxylase $\rightarrow$ more calcitriol $\rightarrow$ restores phosphate. Hyperphosphatemia and FGF23 do the opposite.

BOARD TRAP

This is why phosphate-wasting disorders (FGF23 excess) paradoxically show low/inappropriately-normal 1,25 despite low phosphate — FGF23 override dominates.

12. FGF23 $\uparrow$ CYP24A1 (inactivation)

WHAT YOU SEE

FGF23 / CYP24A1 banner

WHAT IT ENCODES

FGF23 SUPPRESSES 1-alpha-hydroxylase AND induces CYP24A1, the 24-hydroxylase that inactivates vitamin D (to 24,25-D). Net: FGF23 lowers calcitriol and wastes phosphate.

BOARD TRAP

Loss-of-function CYP24A1 = idiopathic infantile hypercalcemia / vitamin-D hypersensitivity — calcitriol cannot be cleared, causing hypercalcemia.

13. 24-hydroxylase (CYP24A1) inactivation

WHAT YOU SEE

Degradation pathway icon

WHAT IT ENCODES

CYP24A1 (24-hydroxylase) catabolizes both 25-D and 1,25-D into inactive 24,25 metabolites — the off-switch for vitamin D signaling.

BOARD TRAP

Calcitriol upregulates its OWN CYP24A1 (negative feedback); granulomas LACK this regulation, causing unchecked 1,25 production and hypercalcemia.

14. Granulomatous CYP27B1 — unregulated

WHAT YOU SEE

Lymphoma/granuloma figure

WHAT IT ENCODES

Macrophages in sarcoidosis, TB, lymphoma express EXTRARENAL CYP27B1 that is NOT suppressed by feedback $\rightarrow$ autonomous 1,25-D $\rightarrow$ hypercalcemia and hypercalciuria.

BOARD TRAP

Granulomatous hypercalcemia: HIGH 1,25-D with LOW/suppressed PTH and normal 25-D — opposite of vitamin-D-deficiency labs; treat with glucocorticoids, not just hydration.